

MODDABLE CHESS · VARIANT 98 OF 102

# Toroidal Chess

**8×8**

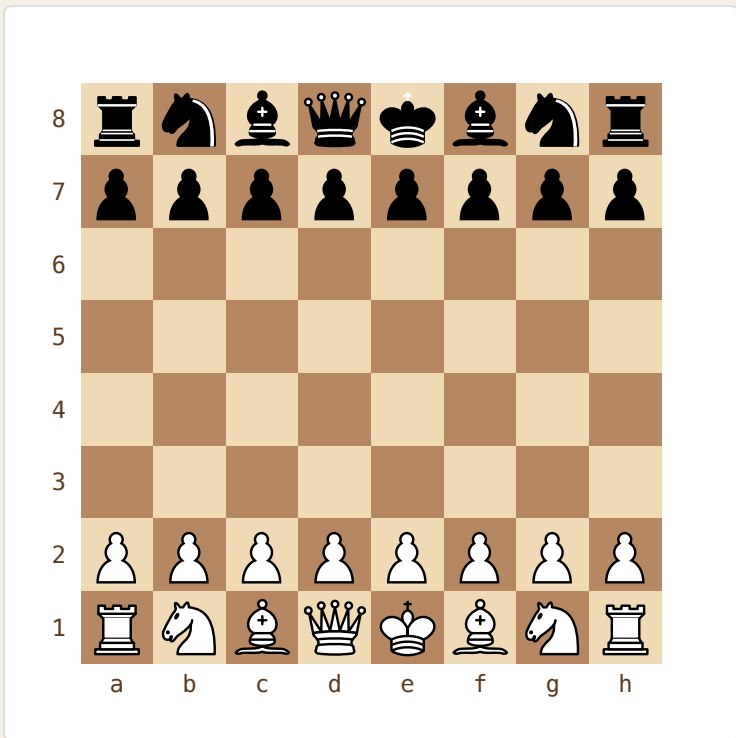
BOARD

**2**

PLAYERS

# TOROIDAL CHESS

The board wraps in both directions: files connect left-to-right and ranks connect top-to-bottom, forming a torus. There are no edges.



*Toroidal Chess – starting position*

## SETUP

**Board:** Standard 8×8 (wraps both axes).

**Setup:** Standard chess setup.

**FEN:** rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR w - - 0 1

## RULES

- ♦ The a-file connects to the h-file (horizontal wrap).
- ♦ Rank 1 connects to rank 8 (vertical wrap).
- ♦ Rooks can slide in any direction and wrap around indefinitely until blocked.
- ♦ Bishops slide diagonally and wrap; a bishop can return to its starting square by sliding far enough.
- ♦ Queens combine both wrapping movements.
- ♦ Knights jump normally but the connectivity means all squares are reachable.
- ♦ No castling (no meaningful concept of king safety with wrapping ranks).
- ♦ No en passant.
- ♦ Pawns move “forward” relative to their colour (White toward rank 8, wrapping to rank 1; Black toward rank 1, wrapping to rank 8).

## WIN CONDITION

Checkmate the opponent's King.

## STRATEGY

On a torus, there are no safe corners or edges; every square is equally central. Bishops can reach every square of their colour (not just 7 diagonals but infinitely wrapping ones). Rooks threaten ranks and files that circle back, making them nearly impossible to evade. The opening concept of “King safety” is meaningless; positional play dominates.

## ATTRIBUTION

Mathematical chess variant based on toroidal geometry. Public domain.